Forecasting, Part 2: Exponential Smoothing

Chapter 16, §16.2 (exponential smoothing, with and without trend)

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Objectives

- Produce an **exponential smoothing** forecast and explain what the smoothing factor α does.
- Produce an **exponential smoothing with trend adjustment** forecast and explain what β does.
- **Tune** α and β on the data, and say why a parameter tuned on the whole sample is not the same as one that would have worked in real time.
- Recognise when a smoothing method is being beaten by a trend, and fix it.

> Before Exam 1 you built forecasts by averaging the last few values. Two things were wrong with that, and you saw both on the CPI. A moving average forgets everything older than its window, and it remembers everything inside the window equally. Exponential smoothing fixes both with one parameter: it remembers *everything*, but with weights that shrink geometrically into the past. Then the trend-adjusted version fixes the lag you watched every average suffer on a rising series.
>
> You have been submitting challenge forecasts for six weeks now. Bring your backtest tables. By the end of this note you should know whether to switch methods, and be able to say why in two sentences.

Exponential Smoothing

Exponential smoothing adjusts the previous forecast by a fraction of the previous period's forecasting error (Formula 16.2):

$$F_t = F_{t-1} + \alpha\,(A_{t-1} - F_{t-1}),$$

where F_t is the forecast for period t, A_{t-1} the actual value in period $t-1$, and α the **smoothing factor**, $0 \le \alpha \le 1$.

Read the formula as a sentence: *next forecast = last forecast, corrected by a share of how wrong it was.* If last quarter's forecast was too low (positive error), the new forecast goes up; too high, it comes down. Donnelly calls it "self-correcting". The size of the correction is α.

> Do not confuse this α with the α of a hypothesis test. Same letter, no relation. Forecasters and hypothesis testers both got there first.

Priming the pump

To forecast period 13 you need the forecast for period 12, which needs the forecast for period 11, and so on back to period 1 — which has no earlier period. The convention is to set $F_1 = A_1$ and start. That first "forecast" was assigned, not calculated, so it gets no forecasting error and it is left *out* of the MAD. Leaving it in would hand the method a free zero and make the comparison with the moving averages unfair.

With Brandywine Ford's sales and $\alpha = 0.6$:

$$\begin{aligned}
F_1 &= A_1 = 97 \\
F_2 &= F_1 + 0.6\,(A_1 - F_1) = 97 + 0.6\,(97 - 97) = 97.0 \\
F_3 &= F_2 + 0.6\,(A_2 - F_2) = 97.0 + 0.6\,(142 - 97.0) = 124.0 \\
F_4 &= F_3 + 0.6\,(A_3 - F_3) = 124.0 + 0.6\,(108 - 124.0) = 114.4
\end{aligned}$$

Notice F_4: the period-3 error was $108 - 124 = -16$, so the forecast came *down* by $0.6 \times 16 = 9.6$. That is the self-correction.

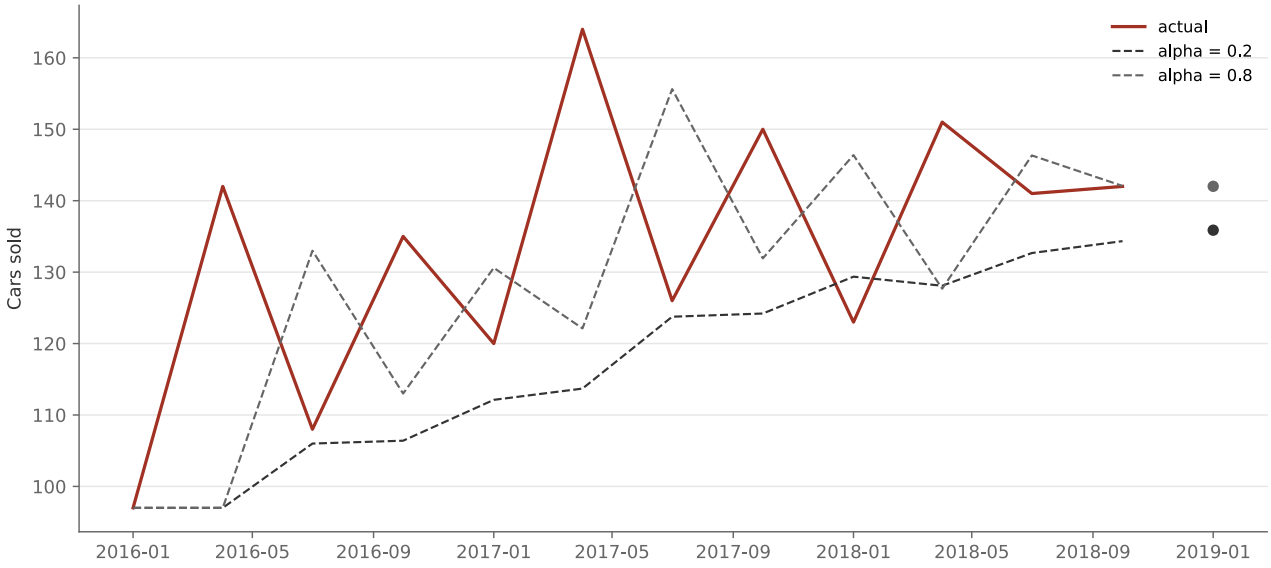
	actual	forecast	error	abs_error
2016Q1	97.0	NaN	NaN	NaN
2016Q2	142.0	97.0	45.0	45.0
2016Q3	108.0	124.0	−16.0	16.0
2016Q4	135.0	114.4	20.6	20.6
2017Q1	120.0	126.8	−6.8	6.8
2017Q2	164.0	122.7	41.3	41.3
2017Q3	126.0	147.5	−21.5	21.5
2017Q4	150.0	134.6	15.4	15.4
2018Q1	123.0	143.8	−20.8	20.8
2018Q2	151.0	131.3	19.7	19.7
2018Q3	141.0	143.1	−2.1	2.1
2018Q4	142.0	141.9	0.1	0.1

The forecast for 2019Q1 is 141.9 cars and the MAD is 19.0 — averaged over eleven errors, periods 2 through 12. That is *worse* than every moving average from Part 1. Donnelly's point, and it bears repeating: no method wins on every series. With this α on this series, exponential smoothing loses.

What α does

A large α carries most of the last error forward: the forecast chases the data and jumps around. A small α barely reacts: the forecast is smooth and slow. The role is the same as the window length p in a moving average, and the right value depends on the data in the same way.

Figure 16.6: the effect of alpha on the forecast

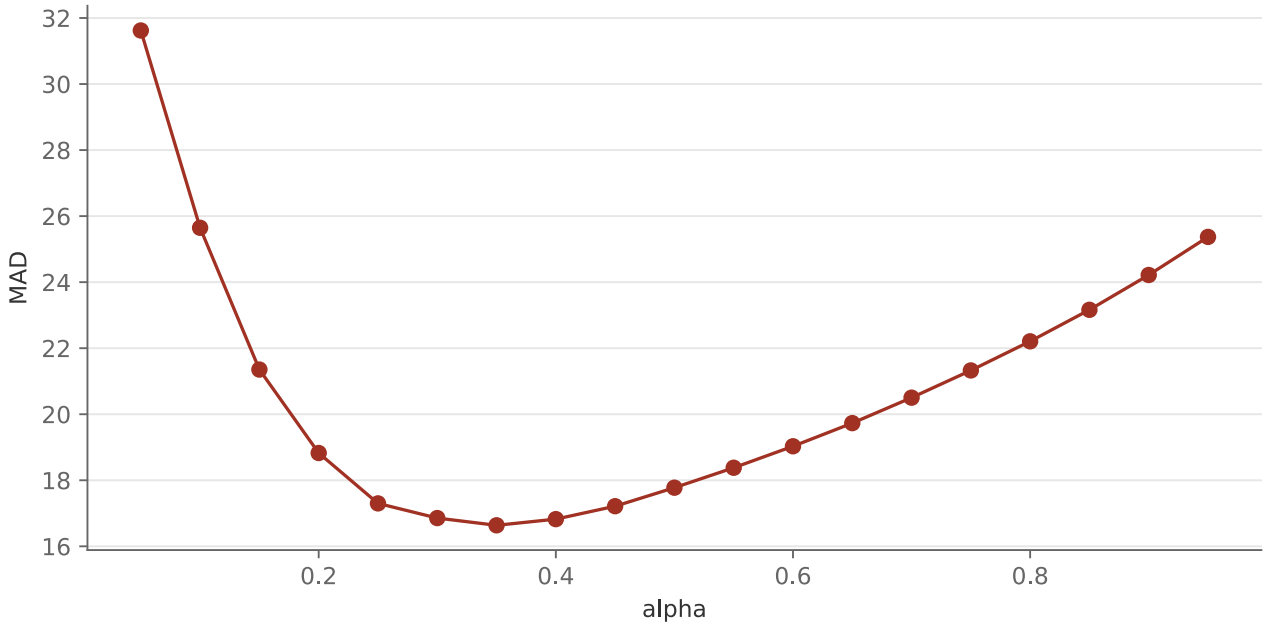


Why "exponential"? Unwind the formula once: $F_t = \alpha A_{t-1} + (1-\alpha)F_{t-1}$. Unwind it again and F_{t-1} contains αA_{t-2} weighted by $(1-\alpha)$; unwind again and A_{t-3} carries $(1-\alpha)^2$. Every past value is in the forecast, with weights $\alpha, \alpha(1-\alpha), \alpha(1-\alpha)^2, \ldots$ — a weighted moving average over the *whole* history, with weights that decay exponentially. With $\alpha = 0.6$ the value from four periods ago has weight 0.038. With $\alpha = 0.1$ it still has weight 0.073.

Tuning α

Since α is a choice, choose it the way you chose p: compute the MAD for a range of values and look.

Figure 16.5: the MAD is U-shaped in alpha



The MAD is smallest at $\alpha = 0.35$, where it is 16.6 — the book eyeballs "around 0.35". Even at its best, exponential smoothing does not catch the four-period moving average on this series (11.2). The seasonal pattern is the reason, and nothing in this note fixes it; that is Part 3.

> A tuned parameter is a *description of the past*. The α that minimises the MAD over 2022–2026 is the one that would have done best *if you had known it in 2022*, which you did not. Below we check whether a tuned α still wins when it is chosen only from data available at the time. When it does not, the tuning has fitted noise. This is the single most useful habit you can take from this course into any job that involves a model.

Excel, for the record

Excel's Data Analysis tool asks for a **damping factor**, not α. The damping factor is $1 - \alpha$: enter 0.4 to get $\alpha = 0.6$. The output column is offset by one row and begins with `#N/A` — that is period 1 having no forecast, which is correct. If you ask Claude to build the workbook, say "damping factor 0.4" and check that the first numeric cell is 97.

Exponential Smoothing with Trend Adjustment

Every method so far lags a trending series, because every one of them forecasts from a level and a trend keeps moving the level. The fix is to smooth two things: the level, and the *change* in the level.

> **Exponential smoothing with trend adjustment** (Formulas 16.3–16.5):
>
> $$\begin{aligned} F_t &= \mathrm{FIT}_{t-1} + \alpha\,(A_{t-1} - \mathrm{FIT}_{t-1}) && \text{the smoothed level} \\ T_t &= \beta\,(F_t - F_{t-1}) + (1-\beta)\,T_{t-1} && \text{the smoothed trend} \\ \mathrm{FIT}_t &= F_t + T_t && \text{the forecast including trend} \end{aligned}$$
>
> with β, the trend smoothing factor, also between 0 and 1. Start with $F_1 = A_1, T_1 = 0$.

T_t is a weighted average of the *most recent* change in the level ($F_t - F_{t-1}$, weighted β) and the *accumulated* trend so far (T_{t-1}, weighted $1 - \beta$). A large β believes the latest change; a small β believes the long-run drift. Then the forecast is the level plus one period's worth of trend.

With $\alpha = 0.3$ and $\beta = 0.1$ (Table 16.6):

	actual	F	T	FIT	abs_error
2016Q1	97.0	97.0	0.0	97.0	NaN
2016Q2	142.0	97.0	0.0	97.0	45.0
2016Q3	108.0	110.5	1.4	111.8	3.8
2016Q4	135.0	110.7	1.2	111.9	23.1
2017Q1	120.0	118.9	1.9	120.8	0.8
2017Q2	164.0	120.5	1.9	122.4	41.6
2017Q3	126.0	134.9	3.1	138.1	12.1
2017Q4	150.0	134.4	2.8	137.2	12.8
2018Q1	123.0	141.1	3.2	144.2	21.2
2018Q2	151.0	137.9	2.5	140.4	10.6
2018Q3	141.0	143.6	2.9	146.4	5.4
2018Q4	142.0	144.8	2.7	147.5	5.5
2019Q1	NaN	145.8	2.5	148.4	NaN

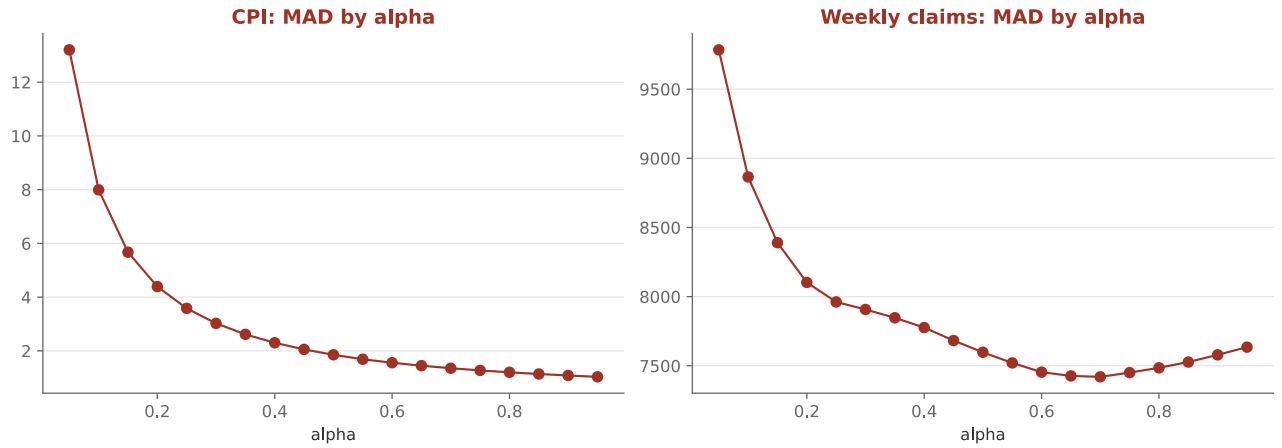
Follow one row. For period 3: the level $F_3 = 97.0 + 0.3(142 - 97.0) = 110.5$; the trend $T_3 = 0.1(110.5 - 97.0) + 0.9(0) = 1.4$; the forecast $\mathrm{FIT}_3 = 111.9$. The last row is the forecast for 2019Q1: 148.4 cars, MAD 16.5. Better than plain exponential smoothing at $\alpha = 0.6$, still not better than the four-period average. The trend is real but small (about 2.5 cars a quarter), and the seasonal swings are ten times larger.

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	142.0	26.6	29.7	19.7	11
SMA p=3	144.7	15.9	19.9	11.2	9
SMA p=4	139.2	11.2	16.0	7.6	8
WMA 3-2-1	143.2	15.3	19.2	10.8	9
ES alpha=0.6	141.9	19.0	23.3	13.8	11
ES+trend 0.3, 0.1	148.4	16.5	21.8	11.7	11

This is Donnelly's Table 16.7, with the naive row added and two extra yardsticks. Read it as the book does: choose the four-period SMA, and do not conclude that simple always beats sophisticated. It depends on the series, and now we go to the series that matter.

The Challenge Series

The CPI: tune, then check the tuning



Two very different curves. The CPI wants the largest α we tried (0.95): a trending series wants to chase the level, and the more it chases, the closer it gets to the naive forecast. Claims settle at α = 0.70: noisy and flat, so some averaging helps, but not much. The parameter is describing the series.

Now the trend component. A grid over α and β for the CPI:

alpha \ beta	0.100	0.300	0.500	0.700	0.900
0.100	3.009	2.525	2.196	2.080	2.154
0.300	1.144	1.125	1.121	1.127	1.144
0.500	0.853	0.878	0.878	0.872	0.857
0.700	0.726	0.750	0.736	0.728	0.724
0.900	0.663	0.659	0.671	0.673	0.694

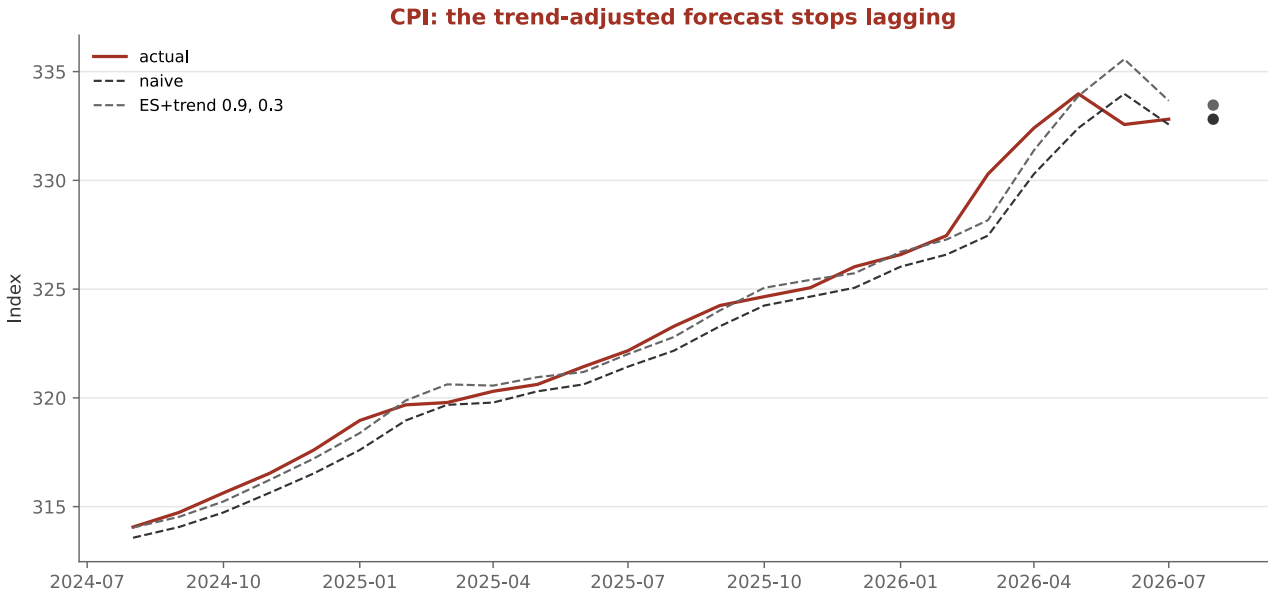
Smallest MAD at α = 0.9, β = 0.3: 0.659, against 0.990 for the naive forecast that won Part 1. A method knows, for the first time, that the CPI drifts upward, and that is what “same as last month” could never know.

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	332.813	0.990	1.253	0.320	54
SMA p=3	333.120	1.821	2.128	0.587	52
ES alpha=0.95	332.804	1.033	1.297	0.334	54
ES+trend 0.9, 0.3	333.461	0.659	0.964	0.214	54

That is the whole-sample verdict for August 2026. Here is the one that counts — the last 24 months replayed one at a time, each forecast made with only the data that existed then:

method	MAD	RMSE	MAPE	periods
ES+trend 0.9, 0.3	0.547	0.862	0.167	24
ES+trend 0.5, 0.3	0.741	1.034	0.227	24
naive	0.919	1.098	0.283	24
ES 0.95	0.955	1.137	0.294	24
ES 0.5	1.617	1.850	0.498	24
SMA p=3	1.647	1.934	0.508	24

Winner: **ES+trend 0.9, 0.3**, with a backtested MAD of 0.547 index points against 0.919 for naive. The tuned parameters survived the backtest, which is the evidence you need before submitting them. Notice also that an *untuned* trend-adjusted model $(0.5, 0.3)$ beats naive too: the gain is from modelling the trend at all, not from the fine tuning.



Weekly claims: no trend to adjust

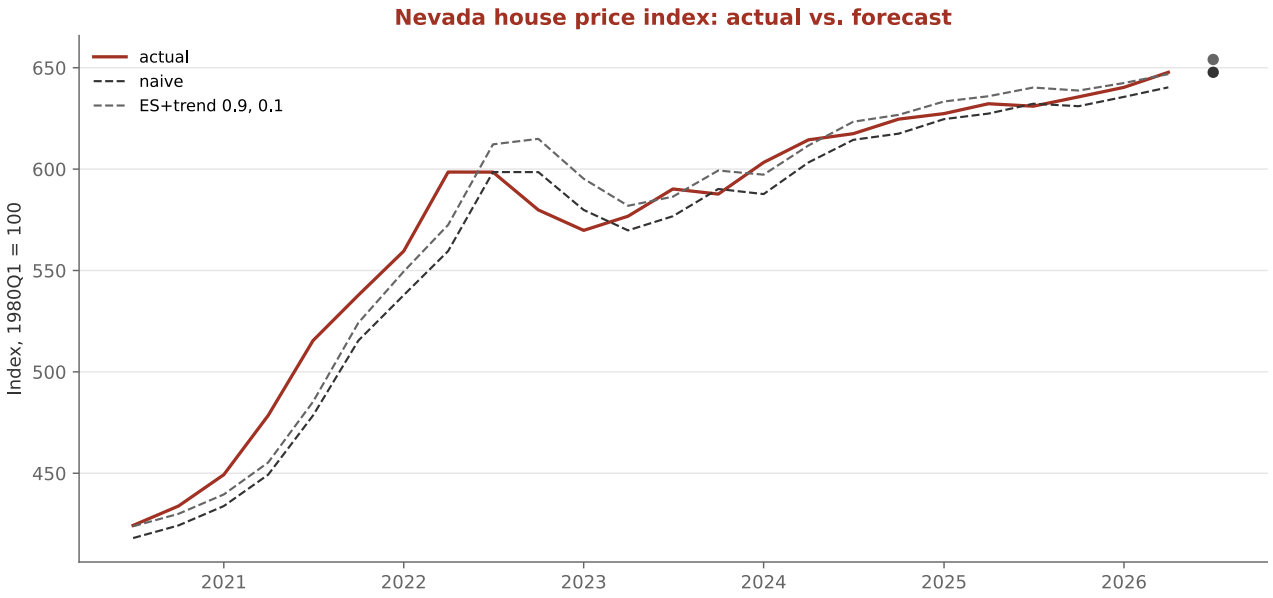
method	MAD	RMSE	MAPE	periods
ES 0.70	6,849.8	9,346.8	3.3	26
SMA p=2	7,019.2	9,655.1	3.4	26
naive	7,692.3	9,762.6	3.7	26
ES+trend 0.3, 0.1	8,111.1	10,441.7	3.9	26
SMA p=4	8,375.0	10,327.2	4.0	26

Over the last 26 weeks the best is **ES 0.70**. The trend-adjusted model, which rescued the CPI, sits at 8,111 here — it is estimating a trend that does not exist and paying for the noise it picks up doing so. A component you model that is not there costs accuracy, exactly as one you fail to model does.

Nevada house prices: quarterly, trending, and turning

method	next_forecast	MAD	RMSE	MAPE	n_errors
naive	647.75	9.87	13.01	2.17	45
SMA p=2	644.04	14.38	18.44	3.13	44
ES 0.95	647.37	10.29	13.51	2.26	45
ES+trend 0.5, 0.3	650.41	10.55	15.92	2.19	45
ES+trend 0.9, 0.1	654.07	7.79	11.16	1.65	45

method	MAD	RMSE	MAPE	periods
ES+trend 0.5, 0.3	3.42	3.94	0.54	8
ES+trend 0.9, 0.1	4.13	4.85	0.66	8
naive	4.47	4.89	0.70	8
ES 0.95	4.65	5.08	0.73	8
SMA p=2	6.48	6.98	1.03	8



The whole-sample winner is **ES+trend 0.9, 0.1**; the backtest winner is **ES+trend 0.5, 0.3**. When those two differ, the disagreement is information: the index rose steeply through 2022 and has flattened since, so a model tuned on eleven years has learned the steep part while a model judged on the last eight quarters is being asked about the flat part. A series that has changed character rewards the method closest to now, and the backtest is the one to believe.

> This is the quarterly series in the challenge. You get two forecasts of it all semester, so there is no learning from mistakes in real time. Everything you know has to come from the backtest.

Working with Claude

The lab prompt

RUN THIS WITH CLAUDE

Get the MBA 775 course materials from https://github.com/jcrooker/mba775. I need `scripts/16b_exponential_smoothing.py`, `scripts/_course.py`, `scripts/_forecast.py`, `scripts/_charts.py`, `data/cpi.csv`, `data/icsa.csv` and `data/nvsthpi.csv`. Read `COURSE_CONTEXT.md` from the same repository first — it explains how this course expects analytical work to be done.

Run the script and show me its actual output, verbatim. Then walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot reach GitHub, tell me plainly and I will upload the files instead.

Update your challenge forecast

By now you have a script from Part 1 and several weeks of published values. This prompt adds the new methods without throwing away the old comparison:

PROMPT — ADD SMOOTHING TO AN EXISTING FORECAST SCRIPT

I have uploaded my forecast script for FRED series ICSA, `_forecast.py`, and a fresh `icsa.csv` downloaded today. First, confirm the last observation date in the new file and tell me how many weeks were added since the file I used before (its last date was).

Then extend the script: keep the naive and moving-average methods as they are, and add exponential smoothing with alpha tuned over 0.05–0.95 using `tune_alpha`, and trend-adjusted smoothing with alpha = 0.3, beta = 0.1. Report the next-week forecast for every method and a `backtest_summary` over the last 26 weeks. Show the code and its verbatim output.

Finally, answer one question from the numbers only: did the tuned alpha do better in the backtest than alpha = 0.3 would have? If it did not, say that I have over-fit and explain what that means in one paragraph.

Explain a formula by tracing it

PROMPT — TRACE FORMULA 16.5

Using the car-sales data in `_forecast.CAR_SALES` and `exp_smooth_trend` with alpha = 0.3 and beta = 0.1, print the F, T and FIT columns for periods 1–5. Then, for period 4 only, show me the three arithmetic steps by hand — the numbers substituted into Formulas 16.4, 16.5 and 16.3 in that order — and confirm each matches the table to one decimal. Do not explain what exponential smoothing is; I know. Show me the arithmetic.

Verify

Check	What it catches
The first forecast equals the first actual ($F_1 = A_1$)	A different priming rule, which changes every number after it
The MAD counts $n - 1$ errors, not n	Period 1's free zero being counted
Excel's damping factor is $1 - \alpha$, not α	The single most common Excel mistake in this chapter
The tuned α was chosen on a stated grid	"Optimal" from a tool you cannot see
The backtest winner is the one you are submitting	A whole-sample winner that fell apart in real time
A tiny β with a large α is nearly naive — say so if that is what won	Sophistication that is doing no work

Your Turn (from the textbook)

Using the GPA series from Part 1 (Donnelly, Your Turn 3 and 4): (a) forecast the senior-fall GPA with exponential smoothing, $\alpha = 0.4$, and its MAD; (b) repeat with trend adjustment, $\alpha = 0.5$, $\beta = 0.7$. By hand first.

method	next_forecast	MAD	RMSE	MAPE	n_errors
ES 0.4	3.033	0.332	0.351	11.598	8
ES+trend 0.5, 0.7	3.212	0.351	0.384	12.737	8

Laboratory 16b

Path 1 — upload the files

Download the Laboratory 16b pack — one file. Unzip it and you will have a folder holding 8 files. Drag all 8 into a new Claude conversation, wait for each to show as attached, then paste this:

> **PROMPT**
>
> I have uploaded 8 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.
>
> Then run `16b_exponential_smoothing.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

> **PROMPT**
>
> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 16b and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

> Then run `16b_exponential_smoothing.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Use Path 1.

Before your next submission, add one line to your index card: the backtested MAD of the method you *stopped* using. If you switched from a moving average to trend-adjusted smoothing this week, you should be able to say by how much, over how many periods, and on which series it would have been a mistake.